TUTORIAL SHEET - 2

(DIGITAL CIRCUITS AND SYSTEMS)

1. (a)

① $F(A,B,C,D) = \pi(1,5,12,15)$

$$F(A,B,C,D) = \sum(0,2,3,4,6,7,8,9,10,11,13)$$

(a)

$$F(A,B,C,D) = I_0\bar{B}\bar{C}\bar{D} + I_1\bar{B}\bar{C}D + I_2\bar{B}C\bar{D} + I_3\bar{B}CD + I_4B\bar{C}\bar{D} + \dots + I_{15}BCD$$

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
	000	001	010	011	100	101	110	111
	$\bar{B}\bar{C}\bar{D}$	$\bar{B}\bar{C}D$	$\bar{B}C\bar{D}$	$\bar{B}CD$	$B\bar{C}\bar{D}$	$B\bar{C}D$	$BC\bar{D}$	BCD
A=0 $\bar{A}$	0	1	2	3	4	5	6	7
A=1 A	8	9	10	11	12	13	14	15
	1	A	1	1	$\bar{A}$	A	1	$\bar{A}$

1. (b)

(b) 2 Variable for select lines &
2 for inputs

B & C → select lines

A & D → input lines

	I_0 00 $\bar{B}\bar{C}$	I_1 01 $\bar{B}C$	I_2 10 $B\bar{C}$	I_3 11 BC
$\bar{A}\bar{D}$	0	2	4	6
$\bar{A}D$	1	3	5	7
$A\bar{D}$	8	10	12	14
AD	9	11	13	15
	$A+\bar{D}$	1	$\bar{D}$	$\bar{A}$

$$\bar{A}\bar{D} + A\bar{D} + AD = A + \bar{D}$$

$$\bar{A}\bar{D} + A\bar{D} = \bar{D}$$

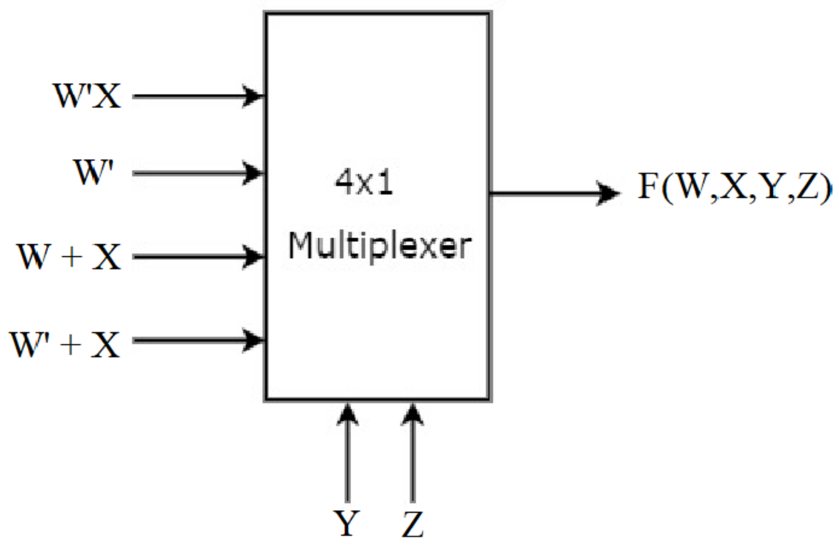
$$\bar{A}\bar{D} + \bar{A}D = \bar{A}$$

2. .

(2.)

$$F(W,X,Y,Z) = \sum m(3,4,5,7,10,14) + \sum d(1,6,15)$$

	I_0 $\bar{Y}\bar{Z}$	I_1 $\bar{Y}Z$	I_2 $Y\bar{Z}$	I_3 YZ
$\bar{W}\bar{X}$	0	1	2	3
$\bar{W}X$	4	5	6	7
$W\bar{X}$	8	9	10	11
WX	12	13	14	15
	$\bar{W}X$	$\bar{W}$ or $\bar{W}X$	$W+X$ or W	$\bar{W}+X$ or $\bar{W}$ → considering don't care.



4.

Truth Table :

BCD Inputs				excess-3 Outputs			
A	*B*	*C*	*D*	*W*	*X*	*Y*	*Z*
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

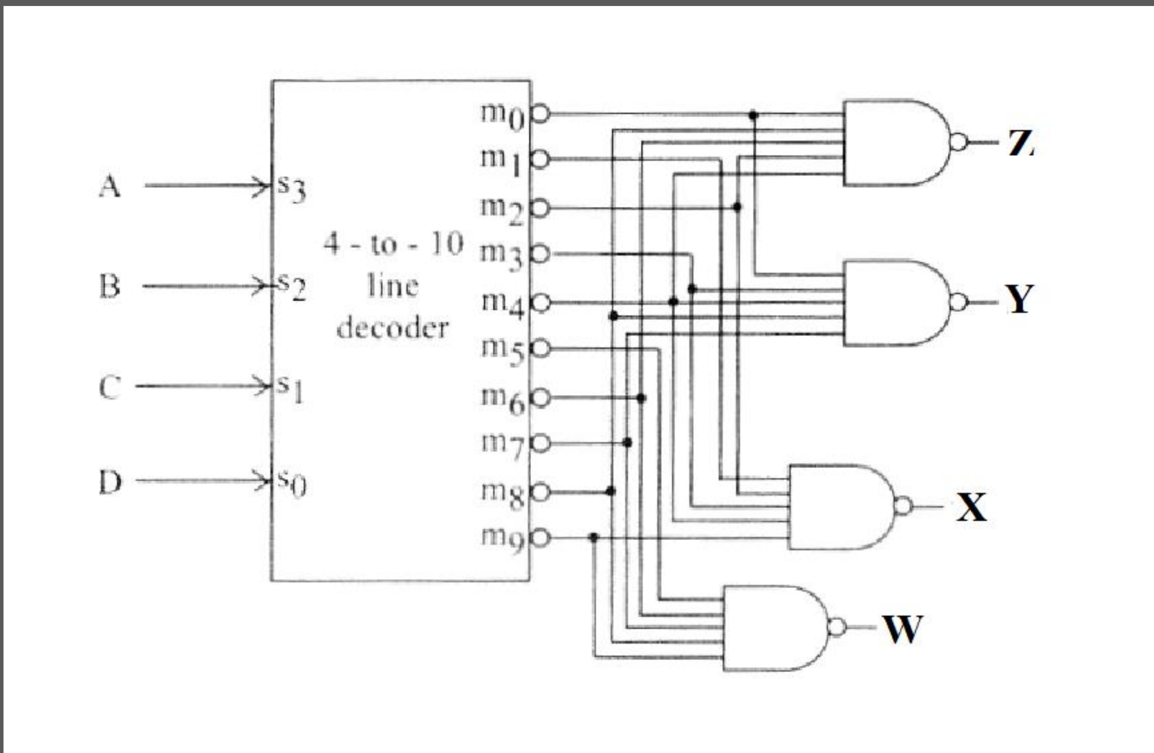
$W = \sum m(5, 6, 7, 8, 9)$

$X = \sum m(1, 2, 3, 4, 9)$

$Y = \sum m(0, 3, 4, 7, 8)$

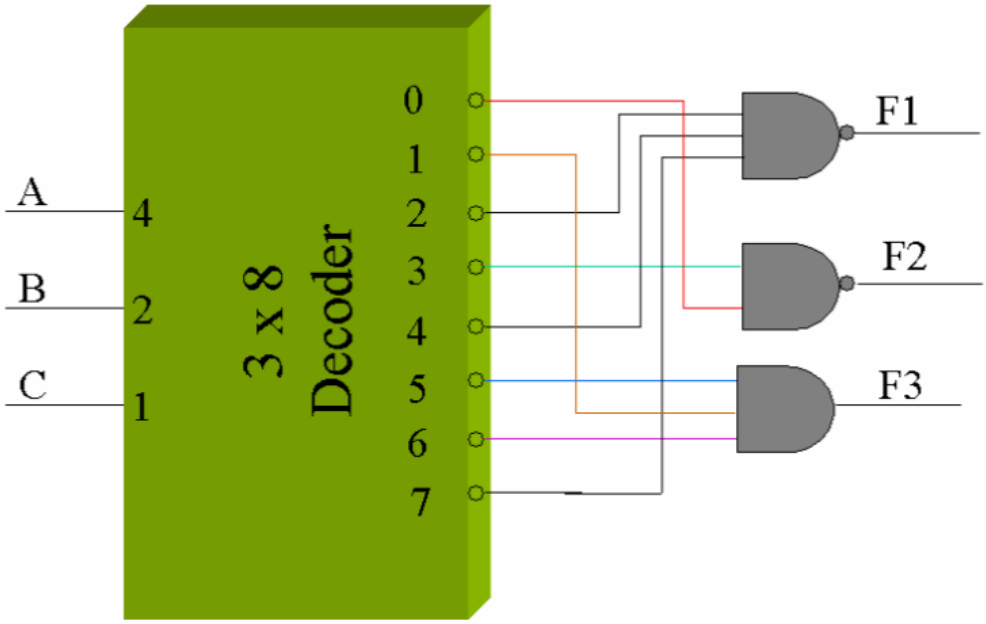
$Z = \sum m(0, 2, 4, 6, 8)$

- Since decoder output are active low, NAND gates are required.
- So four 5-input NAND gates are needed with corresponding to the min-terms of the excess-3 output.



3.

Minimize the number of inputs in the external gates.



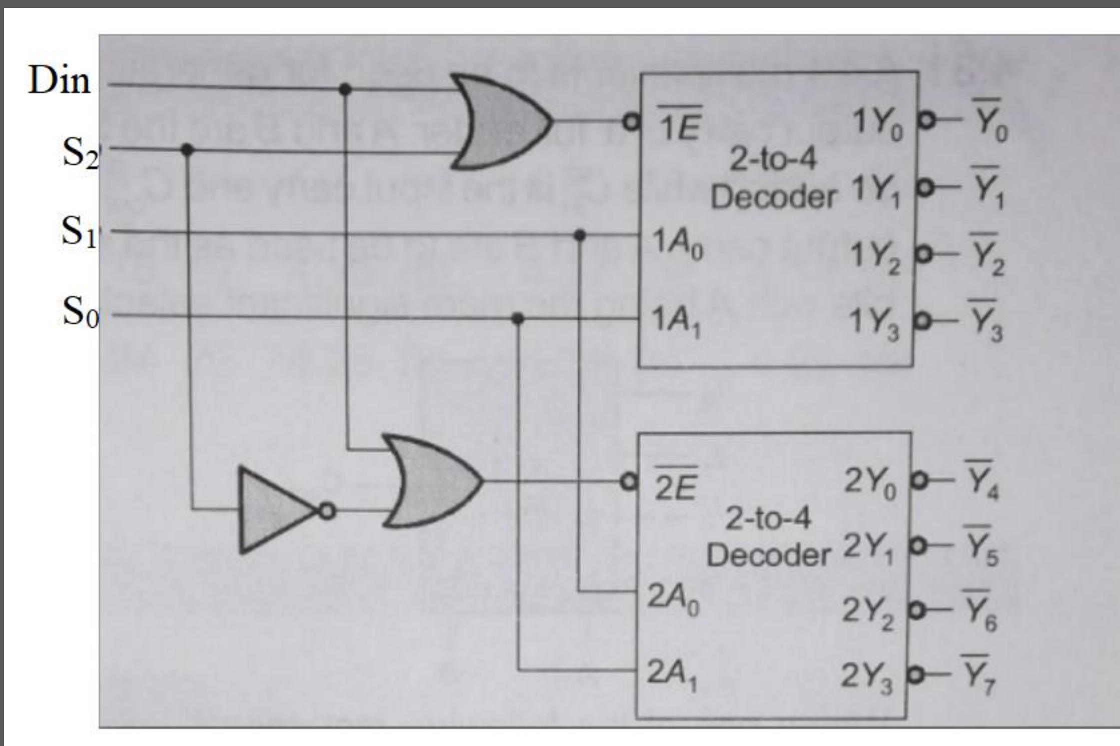
5.

Truth Table of DEMUX :

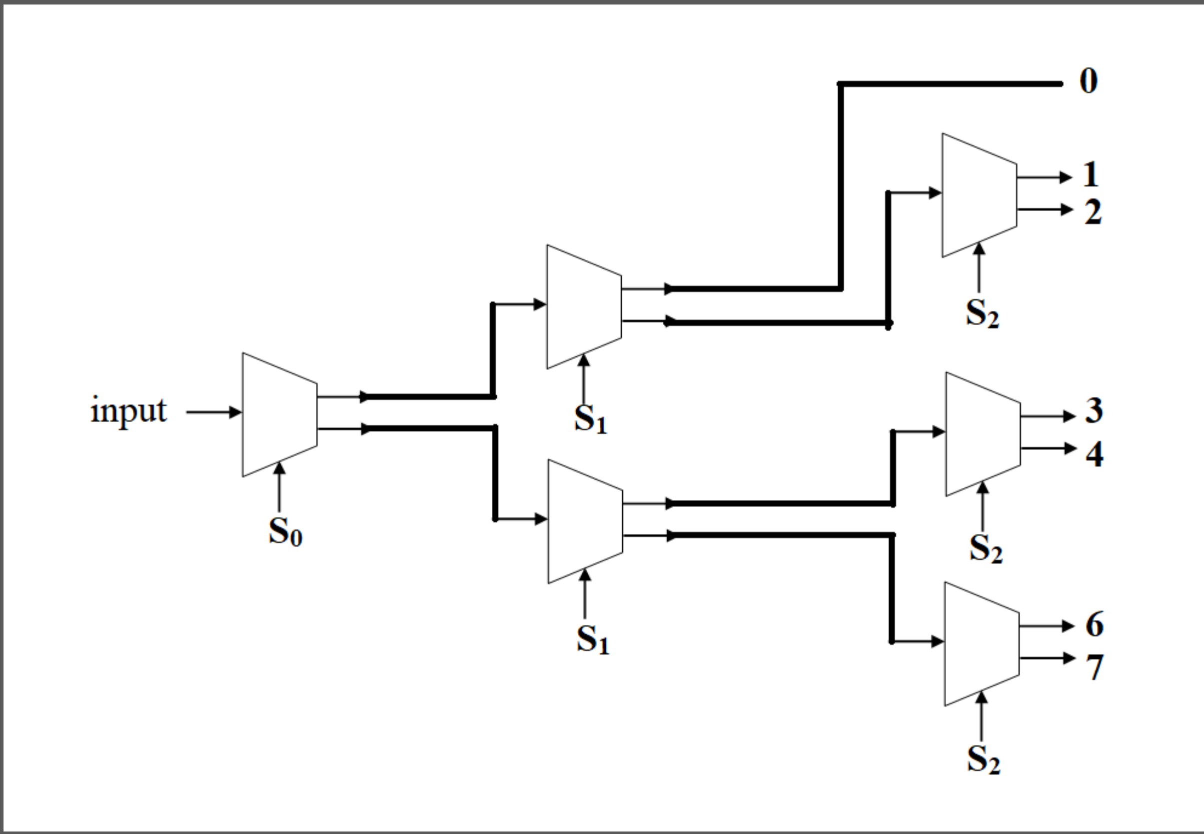
Data Input	Select Inputs			Outputs							
D	S_2	S_1	S_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
D	0	0	0	0	0	0	0	0	0	0	D
D	0	0	1	0	0	0	0	0	0	D	0
D	0	1	0	0	0	0	0	0	D	0	0
D	0	1	1	0	0	0	0	D	0	0	0
D	1	0	0	0	0	0	D	0	0	0	0
D	1	0	1	0	0	D	0	0	0	0	0
D	1	1	0	0	D	0	0	0	0	0	0
D	1	1	1	D	0	0	0	0	0	0	0

Truth Table of decoder :

Inputs			Outputs			
EN	A	B	Y_3	Y_2	Y_1	Y_0
0	×	×	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0



6.



input
S_0
S_1
S_1
S_2
S_2
S_2
0
1
2
3
4
6
7